Commit Now, Walk Away Later
Designing Forfeits for Delayed Blockchain Transactions

CS 6001 Course Project Final Report

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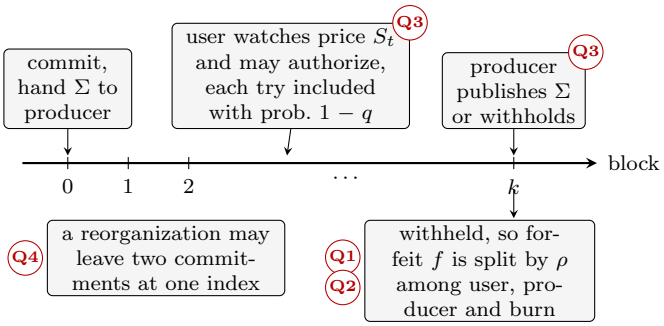
Abstract

A blockchain is a shared record of signed transactions, where digital signatures prove each owner's approval. Standardized quantum resistant signatures are about 38 times larger than today's, so newer blockchain designs propose to keep them off the record and let the user authorize a committed transaction a few blocks after committing. This delay gives the user a free option, since a trade that turns bad can simply be abandoned. Existing penalties are not sized from this option, and they ignore a producer who holds the full signature and can either publish it or collude with the user. We treat this as a mechanism design problem, choosing a forfeit, its destination and a producer fallback that deter speculation, keep both parties on board, limit collusion and keep collateral low. We solve the resulting extensive form game for subgame perfect equilibria and calibrate it to nine years of ether prices. On its own, a forfeit must be 0.39% to 0.74% of a trade over 20 minutes. A credible fallback removes the option, yet collusion pays unless some of the forfeit leaves the user and producer, and sending all of it outside, for example by burning, is cheapest. At 20 minutes, burning 1.35% of the trade keeps their joint gain over the best equilibrium within 0.1 basis points, but a collateral cap of 1% allows windows of at most 11.8 minutes. Within this model, delayed authorization can be made safe, but the feasible window shrinks quickly as markets become more volatile.

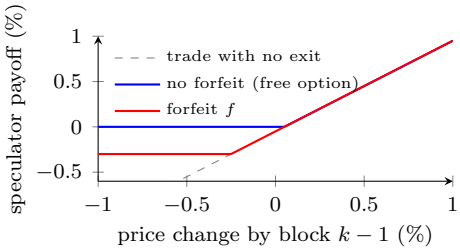
1 Introduction

A blockchain is a shared ledger, a list of transactions of which many independent computers, called nodes, keep identical copies. The ledger grows in batches called blocks, about one every twelve seconds on Ethereum. For each block one participant, the *block producer* (also called a validator or sequencer), chooses which waiting transactions to include and earns a fee for each. Every transaction carries a digital signature, a piece of data that only the account owner can produce and that proves the owner approved the transaction. These signatures are about to grow much larger. A sufficiently large quantum computer would break the signature schemes in use today, and the ML-DSA-44 scheme that NIST standardized as a replacement produces signatures of 2,420 bytes, against 64 bytes for a raw P-256 ECDSA signature [7, 8]. Since every node must receive, check and store every signature, this growth raises the cost of running a ledger for everyone.

One recent proposal avoids this cost by keeping the full signature off the ledger [1]. The user first posts a *commitment*, a short fingerprint (a hash) of the transaction that fixes its contents without revealing them, and a few blocks later posts a short secret value that authorizes it. We call such designs *deferred-authorization protocols*. Their resistance to forgery has been analysed [2], similar delays create free options in atomic swaps, which exchange coins across two blockchains [9], and in Ethereum's planned proposer-builder separation (ePBS), where a separate builder assembles each block [5], and a separate literature studies how producers and users behave when they can make private agreements [21, 22].



(a) Timeline of one commitment



(b) Payoff of walking away

Figure 1. The problem at a glance. Here Σ is the user's full signature, S_t the price at block t, k the deadline block, q the chance that an authorization attempt is left out of a block, f the forfeit charged when the user walks away and ρ the rule that splits it. Red labels mark where each question arises.

Each of these lines of work, however, covers only part of the picture. The security analysis treats producers as adversaries without asking what payments would make them behave well. The swap-pricing literature has no producer who could cancel the option, work on hashed time-lock contracts (HTLCs) [16, 17, 18] protects a two-party exchange rather than one user's delayed authorization, and the ePBS study penalizes the builder, who holds the option itself. The draft EIP-8209 burns, that is destroys, a prepaid fee when a commitment is not revealed [4], and defences against front-running, where others jump ahead of a pending transaction, penalize non-reveal or allow it [6], but neither derives the penalty from the value of the option. Among the works studied, none sizes such a penalty from the option's value and decides where its money should go when the producer can publish the signature and collude with the user.

This gap matters because the delay itself creates the problem. Suppose a user commits to buy an asset for \$100,000 at today's price, for example from a seller whose quoted price stays valid until the window closes, and the price over the next twenty minutes moves with a standard deviation of 0.5%. If the price rises, the user authorizes and gains at the seller's expense, and if it falls, the user stays silent and loses nothing. The user therefore holds a *free option*, the right but not the duty to trade at a fixed price after seeing how the price moved. Staying silent caps the loss at zero, as Figure 1(b) shows for an illustrative fee of 0.05%, and a forfeit f, a penalty charged when the user walks away, lowers that floor to $-f$. This option is worth about $0.4 \times 0.5\%$ of the value, roughly \$200 per commitment, far more than a typical fee.

Two mitigations have been proposed, but each creates a new difficulty. A *forfeit*, a payment charged when a commitment is never authorized, fails if it is too small, and if it is paid to the producer, the producer is tempted to delay an honest user until the window closes so that it can collect the forfeit. A *fallback* lets the producer, which privately receives the user's full signature with the commitment, publish it if the user stays silent, so the large signature reaches the ledger only then. This removes the option, but the user and the producer may then agree in private that the producer stays silent when the price falls and that the two share the gain when it rises. Although the setting comes from blockchains, this is a classical mechanism-design problem: a designer who cannot see types must use payments to deter a profitable deviation, while the agent who enforces the rules has interests of its own and may collude with the user.

The objective of this report is to find forfeit designs that remove the option while staying safe against producers and against collusion. The scope is the incentives around a single commitment, not the bandwidth the design saves. Within this scope it answers four questions, marked in Figure 1(a), the last about the numbered account slot that a commitment occupies, called an *index*.

- **Q1.** How small can a forfeit be and still remove the option, given the window, volatility and size?
- **Q2.** Where should the forfeit go so that the producer still publishes and honest users are unharmed?
- **Q3.** When can a producer, alone or together with the user, profit by keeping the option alive?
- **Q4.** Should a dispute over an index be settled by a bonded contest or by a race?

To answer these questions, the protocol is modelled as a game in which the user and the producer move in turn and a chance player, nature, draws the price at each block. The designer's choice is then written as an optimization over mechanisms with deterrence, individual-rationality, credibility, collusion-proofness and collateral constraints. The game is solved by backward induction, that is, by working out each player's best move from the last block back to the first, with proofs wherever a closed form exists and numerical solutions elsewhere. The report contributes a game model with an independently checked solver, one main analytic result (an impossibility result for forfeits that stay with the user or producer), six shorter propositions, among them the optimality of burning, and calibrated answers to all four questions. Sections 2 to 8 cover related work, the model, the analytic results, the solver and calibration, the answers, a discussion of scope and the conclusion.

2 Related Work

Table 1 lists the closest lines of work, what each establishes and what this report adds.

Burning against collusion, collateral for credible revelation and the window argument are therefore known. The contribution of this report is narrower: it combines them for a producer that holds a usable signature, proves that a forfeit kept within the pair cannot stop collusion (Proposition 4), identifies the cheapest routing (Proposition 7), and computes the windows that a collateral limit allows.

3 Model

Players, timeline and information. The model follows one commitment, as in Figure 1(a). The players are a user U, a block producer P and nature, which is not strategic and stands for the randomness of the price. A block lasts twelve seconds, and the window k is the number of blocks between the commitment and the deadline. At block 0 the user commits and, when the protocol requires it, hands the full signature Σ to the producer. The price S_t at block t follows a recombining binomial lattice, a tree in which the price moves up by the factor $u = e^{\sigma}$ or down by $d = 1/u$ at each block, where σ is the per-block volatility. After j up-moves in t blocks the price is $S_t = S_0 e^{\sigma(2j-t)}$, and an up-move has probability $p = (1-d)/(u-d)$. This makes S_t a martingale, meaning that the expected future price equals the current price. At blocks $1, \ldots, k-1$ the user may attempt to authorize, and each attempt is included with probability $1-q$. If the user has not authorized by block k, a producer holding Σ chooses whether to publish it. If the producer withholds it, or never received it, the commitment is dropped and the user is charged the forfeit f. Every player sees all earlier moves before it acts, so this is an extensive-form game with perfect information and chance moves (Figure 2, whose payoffs are defined in the next paragraph).

Payoffs. Outcomes are measured in money, and players are risk neutral with quasi-linear utilities, so money can be passed between players one for one. The transaction has value N and is priced at the commitment-time price S_0, so executing it at block t is worth $g_t = N(S_t/S_0 - 1)$ to the user. On

Line of work	What it establishes	What this report adds
Game theory for blockchains [27, 28, 29, 30, 31, 32]	Relaying, mining and fee markets are standard game-theoretic questions at top venues	The same tools applied to a delayed authorization
Transaction-fee mechanisms [21, 22, 23, 24]	Fee rules as mechanisms with user, miner and collusion constraints. EIP-1559 resists off-chain agreements (OCA-proofness) only because it burns its base fee	Collusion-proofness here is OCA-proofness for one user and one producer, and Proposition 4 is its analogue for forfeits
Credible auctions with commitments [33, 34, 35, 36]	A bidder who commits but does not reveal pays a fine or loses collateral, sized to make the auction credible	Here the enforcer, the producer, can complete the transaction itself, and where the money goes matters
Deferred authorization [1, 2]	Commit first, authorize later, to keep large signatures off the ledger. A censoring adversary can forge unless eligibility is fixed before disclosure	Adds a fallback, a forfeit and a dispute rule, and asks what a producer *wants* to do rather than what it *can* do
Option value of delay [9, 12, 10, 13, 14, 15, 5]	Atomic swaps and ePBS create free options, priced with the binomial lattice, and penalize the party that holds the option	Here a third party, the producer, can switch the option off or keep it alive by colluding
HTLC incentives and Rapidash [16, 17, 18]	Miners enforce penalties or are bribed. Rapidash burns most funds, pays a small reward, and sets a T-block timeout so that a coalition with mining share γ holds every block only with probability γ^T	Pool routing and Proposition 5 rest on these ideas, applied to a producer that holds a usable signature under a collateral limit
Commit-reveal penalties in practice [4, 6, 3, 19, 20]	Fees for not revealing are burned, pooled or slashed, at amounts fixed by rule	The penalty is sized from the value of the option
Payments and budget balance in mechanism design [43, 44, 45, 46, 47, 48, 49]	Where collected money goes, and its tension with budget balance	The routing question: money must leave the user–producer pair
Producer extraction and disputes [25, 26, 50, 51, 52]	Producers profit from ordering; disputes are settled by bonds or all-pay races	Bond and race compared for disputed indices (Q4); ordering profits left for future work

Table 1. Related work and what this report adds.

execution the user pays the producer a flat fee ϕ, and publishing the signature costs the producer c. Any transaction whose terms are fixed at commitment has this form, such as a purchase at a price that a seller has quoted and must honour until the deadline. An ordinary limit order does not, since it executes at the market price whenever that is better, and a plain transfer creates no option. A speculator commits only for the option, while an honest user values execution at a constant v with $v > \phi$, so its surplus from executing is $v - \phi$. Figure 2 gives the payoff of each outcome, and Table 2 lists the notation and the baseline values. Money is quoted as a fraction of N, and one basis point (bp) is 10^{-4}. A routing rule $\rho = (\rho_U, \rho_P, \rho_O)$ splits a dropped forfeit into the shares returned to the user, paid to the producer and sent outside the pair, with $\rho_U + \rho_P + \rho_O = 1$. Self is $(1, 0, 0)$, burn is $(0, 0, 1)$ and destroys the money, producer is $(0, 1, 0)$, and pool(n) gives the producer $1/n$ and sends the rest outside the pair. The collateral limit κ_c caps the forfeit at $\kappa_c N$.

Threat model. The threats are economic, not cryptographic. Every party acts in its own interest, and the signature and hash commitment are assumed secure [2]. The producer earns only fees and its share of forfeits, and may censor attempts, withhold the signature, or agree with the user on an enforceable side payment. The committed transaction stays valid and executable throughout

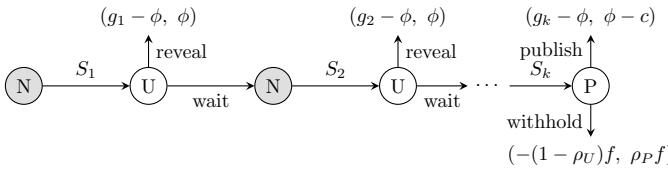


Figure 2. The base game, drawn for $q = 0$ so that every authorization is included, with payoffs (user, producer). Circles marked N are nature's price moves, g_t is the value of executing at block t, ϕ the fee, c the producer's publishing cost, and ρ_U and ρ_P the shares of the forfeit f returned to the user and paid to the producer.

Symbol	Meaning	Baseline value
k	window length in 12-second blocks	100 blocks (20 minutes)
σ_{day}	daily volatility (standard deviation of daily log price changes) of ether	4.57%, interval 4.23%–4.98%
q	probability that an authorization attempt is censored, that is, left out	0, varied in Q1
N, ϕ	transaction value, and flat fee paid to the producer on execution	one unit, 5 bp
c, $v-\phi$	producer's publishing cost, and honest user's surplus	1 bp, 10 bp
$\rho = (\rho_U, \rho_P, \rho_O)$	shares of f to the user, the producer and outside the pair	self, burn, producer, pool(n)
f, κ_c	forfeit for an unauthorized commitment, and collateral limit as a fraction of N	varied, 1%

Table 2. Notation and baseline values.

the window: the user cannot invalidate the fallback by moving funds, changing its transaction counter (nonce) or permissions, or cancelling, and the seller's quote or reserved assets keep the terms enforceable, without which Proposition 3 would fail. Order profits, network attacks and larger coalitions are not modelled.

Solution concept. The game is solved for its subgame-perfect Nash equilibria (SPE) by backward induction. Subgame perfection requires every plan to be a best response at every point of the game, not only on the path played, so it rules out equilibria that rest on threats a player would not carry out. Write V_0 for a user's expected payoff from committing, its value at block 0. A design *deters* speculation if the speculator's V_0 is at most 0 in every pure SPE, where an indifferent producer may break its tie differently at each state. For a strategy profile s, let $J(s)$ be the expected joint payoff of the user and producer, and let J^* be the largest joint payoff they can reach over all joint strategies with side payments. A coalition's *gain* is $\Delta = J^* - \max_s J(s)$, where the maximum runs over SPE with uniform tie-breaking. Under a strictly credible fallback this best equilibrium is the *honest* one, in which the user tries to authorize at every block, but the user is indifferent among authorization times, and in the SPE where it waits for the fallback the joint payoff is lower by $c(1-q^{k-1})$. When the producer prefers to withhold, the best equilibrium already involves strategic timing. Collusion guarantees are thus relative to the best equilibrium, not to every one.

The designer's problem. The question is one of mechanism design with transfers. A designer, whoever writes the protocol rules, fixes a mechanism $M = (k, f, \rho, \mathcal{A})$ before play, where $\mathcal{A}$ states whether the producer may publish the signature and whether the user must hand it over. The designer cannot see who is a speculator and cannot dictate actions, so, as with the payment rule of a classical mechanism, it acts only through the forfeit and where its money goes. M induces the extensive-form game above, an indirect and dynamic mechanism, and the aim is to implement in SPE an outcome in which speculators gain nothing by entering and honest users still want to participate [37]. Write V_0^{spec} and V_0^{hon} for the values of committing to a speculator and to an honest

user. With a collusion tolerance $\varepsilon \geq 0$ and the collateral limit κ_c, the designer's program is

$$\min_{M} \; f \quad \text{subject to} \quad \begin{array}{ll} \text{(D)} \;\; V_0^{\text{spec}}(M) \leq 0 \text{ in every pure SPE} & \text{deterrence} \\ \text{(IR)} \;\; V_0^{\text{hon}}(M) \geq 0 & \text{individual rationality} \\ \text{(CR)} \;\; \phi - c > \rho_P f \;\; \text{(designs with a fallback)} & \text{credibility} \\ \text{(CP}_\varepsilon\text{)} \;\; \Delta(M) \leq \varepsilon & \text{collusion-proofness} \\ \text{(K)} \;\; f \leq \kappa_c N & \text{collateral.} \end{array}$$

The objective is the collateral an honest user must lock. (D) is weak deterrence, a speculator indifferent at zero being assumed to abstain. (IR) guarantees participation, not the full surplus $v-\phi$ (about 0.18 bp short at $k = 10$ and $q = 0.6$ without a fallback). (CR) is the incentive constraint of the producer, the agent that enforces the rules, much as a credible auction needs an auctioneer who prefers to follow them [33]. (CP_0), the case $\varepsilon = 0$, is OCA-proofness for one pair [21] and plays the role of the coalition incentive constraint of collusion-proof mechanism design [38]. The routing ρ decides budget balance. A forfeit kept inside the pair ($\rho_O = 0$) balances the pair's budget, while burning destroys money, as in money-burning mechanisms [39]. The questions solve restricted versions of this program: collateral is minimized with the window fixed, the largest feasible window is then searched for separately, and a credible fallback already meets (D) at $f = 0$. Q1 fixes the window, rules out a fallback and finds the smallest burned forfeit meeting (D). Q2 asks which routings meet (CR) and (CP) together. Q3 fixes the window, requires a credible fallback, finds the cheapest routing and forfeit meeting (CP_ε), and then finds the largest window for which that forfeit meets (K). Q4 compares two rules for a related dispute. Appendix B gives each modelling choice its reason, the alternative not taken and what could come next.

Extensions. Three extensions relax the base game. In the *rotating-producer* model each block is produced, independently, by a strategic entity with probability α (its share of the blocks) and otherwise by an honest producer, so $\alpha = 1$ is a single sequencer, the sole producer of a rollup.

In the *contest* model, a reorganization, in which the latest blocks are replaced, leaves two valid commitments at one index. An attacker with value V_a and cost κ rebinds the index, and the owner, who would lose V_o, is present with probability a. Under a *bond* the attacker posts b, and a present owner with capital $K \geq b$ answers at cost c_r and wins it. Under a *race* both sides pay for priority, an all-pay contest [52].

In the *private-type* model, each user has a hidden exposure θ that makes execution worth θN more, and the designer picks the forfeit that best balances the gain exposed users make by timing their authorization, a chosen proxy for value taken from counterparties, against surplus lost by deterred users (Appendix D). A menu of (window, forfeit) pairs, in the spirit of the revelation principle [41], is an open extension.

4 Analytic Results

Seven results explain most of the numbers. Proposition 4 is the main new result, Propositions 1 to 3 are short lemmas, Propositions 5 and 6 adapt known arguments, and Proposition 7 identifies the cheapest routing. The last three are stated in Appendix A, where all proofs are written step by step. Propositions 2 to 4 rest on the martingale property $\mathbb{E}[g_{t+1} \mid S_t] = g_t$, where $\mathbb{E}$ denotes expectation, that is, the trade's expected value one block later equals its value now. We write $x^+ = \max(x, 0)$. The first result disposes of the simplest routing.

Proposition 1 (Self-routing is inert)**.** *If the forfeit returns to the user, no payoff depends on f.*

Proof. The user's payoff on a dropped commitment is $-(1-\rho_U)f = 0$, and no other payoff involves f. □

The next result measures the option that remains when there is no fallback, and so answers Q1 in closed form.

Proposition 2 (Value of the option)**.** *Without a fallback, with $k \geq 2$ and $q = 0$, let the user lose f when the commitment is dropped. Then the speculator's value is*

$$V_0(f) = \mathbb{E}[\max(g_{k-1} - \phi, -f)] = -f + \mathbb{E}\big[(g_{k-1} - (\phi - f))^+\big] = \mathbb{E}\big[((\phi - f) - g_{k-1})^+\big] - \phi.$$

Speculation is deterred if and only if $\mathbb{E}[(g_{k-1} - (\phi - f))^+] \leq f$. V_0 is continuous and non-increasing in f and equals $-\phi \leq 0$ once $f \geq \phi - \min g_{k-1}$, so a smallest deterring forfeit f^* exists whenever $V_0(0) > 0$.

The proof (Appendix A) shows that waiting is always worth at least authorizing, because $\max(x - \phi, -f)$ is convex and the price is a martingale; $k \geq 2$ is needed because with $k = 1$ there is no block at which to authorize.

In words, the speculator should wait until the last block and authorize only if the trade is then worth more than the fee minus the forfeit. Because a dropped commitment is always observed, the deterrence condition is the analogue of Becker's classical rule for penalties with certain detection, that the penalty must cover the expected gain from the offence [42], with the gain here being the value of an option. Ruling out early authorization extends Merton's result that an American call is never exercised early [11]. The third form says that f^* is the forfeit at which a put option on g_{k-1} with strike $\phi - f$, a contract paying the amount by which g_{k-1} ends below $\phi - f$, is worth exactly the fee ϕ. The next result gives the condition under which the producer uses the fallback.

Proposition 3 (Credibility threshold)**.** *With a fallback, no contestation and the signature handed over, the producer publishes at block k if and only if $\phi - c > \rho_P f$, and it is indifferent at equality. If it publishes, then $V_0 = -\phi$ for every $q \in [0, 1]$ and every forfeit. If it withholds, the game is the one without a fallback.*

A credible fallback thus removes the option for any forfeit. The next result shows that the user and producer can nevertheless undo it whenever the forfeit stays inside the pair.

Proposition 4 (Forfeits that stay inside the pair)**.** *Let there be a fallback with the signature handed over, $k \geq 2$, $\sigma > 0$, $q < 1$ and no contestation, and let $\rho_O = 0$. If the producer strictly prefers to publish, that is $\phi - c > \rho_P f$, then every equilibrium has a joint payoff of the user and producer of at most 0, while a coalition with side payments can obtain at least $(1-q)\mathbb{E}[g_1^+] > 0$. Such a design is therefore not collusion-proof.*

Proof idea. Under strict credibility every commitment executes, the fee cancels inside the pair, and optional stopping caps the joint payoff of any equilibrium at 0. The coalition instead authorizes at block 1 only after a price rise and lets the producer withhold otherwise, which costs the pair nothing because $\rho_O = 0$. The full proof is in Appendix A.

Strictness is needed, since at equality the outcome depends on tie-breaking. In mechanism-design terms, a forfeit that balances the budget of the user–producer pair cannot be both credible and collusion-proof, so some money must leave the pair. This is balance within the pair, not budget balance of a whole mechanism, so the result is an analogue of, not a derivation from, the classical tension between budget balance and other goals [46, 40]; it is closest to Roughgarden's case for burning the base fee.

A producer that receives part of the forfeit can also harm an honest user directly. It can refuse to include the user's authorization until the window closes and then withhold the signature, collecting the forfeit, which this report calls extortion. Sharing block production among many producers

k	f^*	daily vol.	f^*	size	f^*
10	0.083%	1%	0.044%	0.1	0.064%
25	0.191%	2%	0.157%	1	0.551%
50	0.329%	4%	0.456%	10	1.048%
100	0.551%	6%	0.807%	100	1.443%
200	0.897%	8%	1.192%		
400	1.425%				

Table 3. f^*/N against window k (blocks), daily volatility and transaction size (a multiple of the baseline, with the fee fixed in money).

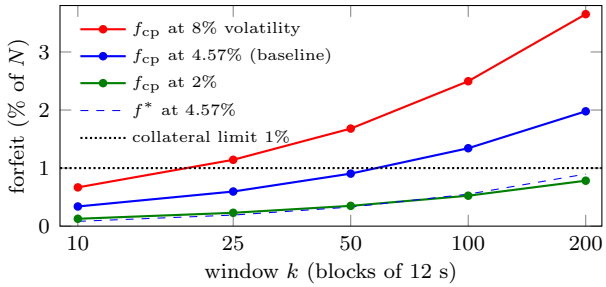


Figure 3. Feasibility of a burned forfeit with a fallback. f^* is the smallest deterring forfeit without a fallback (Proposition 2), and f_{cp} the smallest burned forfeit that keeps the coalition gain within 0.1 bp (Section 6).

makes extortion much harder.

Proposition 5 (Appendix A) makes this precise for the rotating-producer model with $q = 0$. An honest user loses nothing if $\alpha^{k-1}\rho_P f < \phi$ and strictly loses if $\alpha^{k-1}\rho_P f > \phi$, where α^{k-1} is the chance that the strategic entity produces every authorization block, as in Rapidash's timeout argument [18].

5 Experimental Framework

Implementation and verification. The propositions cover the cases with closed forms, and the rest come from a lattice solver, written in Python with NumPy and SciPy (`https://github.com/reman-krishnan/cs6001-project`), that computes equilibria by backward induction over the price tree. It reproduces a hand-computed game, agrees with a separate game-tree solver, exhaustive enumeration and textbook binomial option prices, and passes all 196 automated tests, which catch all 12 deliberately injected bugs (Table 6 in the appendix).

Calibration. Volatility is the root-mean-square daily log return of Binance ETH/USDT closes, 17 August 2017 to 30 September 2026, giving $\sigma_{\text{day}} = 4.57\%$ from 3,331 returns with a 95% moving-block bootstrap interval of 4.23%–4.98%. The per-block volatility of Section 3 is $\sigma = \sigma_{\text{day}}\sqrt{12/86400}$, which assumes uncorrelated log returns. The other values in Table 2 are chosen and varied below, and displayed minimum forfeits and bonds are rounded up.

6 Results

6.1 Q1. The smallest deterring forfeit

The first question is how large a forfeit must be when it is the only defence. Table 3 answers it for a burned forfeit, no fallback and a flat fee of 5 bp, varying one factor at a time. At the baseline, f^* is 0.551% of the transaction value (interval 0.49%–0.63%), and it grows with the window because prices have more time to move. An 8-fold rise in volatility raises f^* 27-fold, because by Proposition 2 the fixed fee then offsets a smaller share of the option, and for the same reason a flat fee covers less of the option on a large trade.

A Gaussian approximation of this put (Appendix D) stays within 0.9% of the lattice for fees from 1 to 20 bp. Censoring matters little because retries absorb it (f^* is 0.475% even at $q = 0.9$). Market conditions matter more: volatility across four sub-periods ranges from 3.53% to 5.61%, so f^* ranges from 0.39% to 0.74%.

Cost to honest users and feasibility. An honest user who retries pays only if all $k-1$ attempts are censored, with probability about 1.1×10^{-22} at $q = 0.6$. The forfeit must, however, be locked as collateral, and under $\kappa_c = 1\%$ only 25 of 40 window and size combinations are feasible.

6.2 Q2. Where should the forfeit go?

Q2 asks whether a fallback does better than a forfeit alone, and where the forfeit's money should go. Over the pooled period (Figure 4 and Table 7 in the appendix), a forfeit alone deters only above f^*, while a fallback with a burned forfeit deters at any forfeit ($0.3f^*$ shown), leaving the speculator only the fee loss $-\phi$ that Proposition 3 predicts. Producer routing fails because the producer stops publishing, and an optional signature and self routing fail too, in all five periods.

6.3 Q3. Producers and coalitions

By Proposition 3, a credible fallback deters a lone speculator even with no forfeit, so the forfeit's job becomes keeping the producer from exploiting the user and the two from colluding. Q3 studies both.

Extortion by a single producer. A single sequencer that decides every inclusion censors every attempt and then withholds if and only if $f > \phi/\rho_P$. An honest user then loses its surplus and the forfeit once f exceeds 0.050% under producer routing, and 0.200% under pool(4), while under burn or self routing the producer gains nothing from a drop. Producer routing is thus doubly weak: it fails Q2, and f^* is eleven times its extortion threshold.

Rotating producers. When a strategic entity produces only a share α of the blocks, the extortion threshold grows by $\alpha^{-(k-1)}$ (Proposition 5, confirmed by the solver; 2,048 at $\alpha = 0.5$ and $k = 12$), so extortion mainly threatens a rollup with one sequencer.

Coalitions. Table 18 gives the coalition gain Δ of Section 3 in a fallback design. Under burn it falls from 10.4 bp at $0.5f^*$ to 0.013 bp at $3f^*$. Under self routing it stays at 21.4 bp whatever the forfeit, because the forfeit is only a transfer inside the pair, and under producer routing it rises with the forfeit, because the producer stops publishing. Proposition 4 explains this pattern, and all 710 random designs meeting its hypotheses have a positive gain. Under burn the benchmark is the honest equilibrium. At a burned forfeit of 1.35% the gain is 0.095 bp over it but 1.095 bp over the equilibrium in which the user waits for the fallback, since at $q = 0$ the pair then also saves $c = 1$ bp. A guarantee across all equilibria at $q = 0$ would need $\Delta + c \le \varepsilon$, so a publishing cost below the tolerance is necessary but not sufficient, and the forfeit would have to be recomputed for the budget $\varepsilon - c$.

How large must a burned forfeit be? Let f_{cp} be the smallest burned forfeit that keeps the coalition gain Δ within the tolerance ε. Proposition 4 rules out only routings that keep the whole forfeit inside the pair, and Proposition 7 (Appendix A) settles which of the rest is cheapest. With a strictly credible fallback the coalition gain depends on the routing only through the amount $\rho_O f$ that leaves the pair, so burning everything needs the smallest forfeit, and pool(n) with $n > 1$ needs $f_{\mathrm{cp}}/(1 - 1/n)$, which is credible only for $n > 1 + f_{\mathrm{cp}}/(\phi - c)$. At the baseline this gives $n \ge 35$ and 1.39%, above the 1% collateral limit, matching the solver, while pool(4) fails because withholding pays its producer. Proposition 7 also shows that a burned forfeit of at least $c - \min g_k$, the publishing cost plus the largest price fall the lattice allows, gives exact collusion-proofness, but at the baseline even a gain within 0.001 bp needs 1.99%, about twice the collateral limit (Table 17). The design therefore aims at (CP_ε) with $\varepsilon = 0.1$ bp, for which f_{cp} is 1.35% at the baseline. On a \$100,000 trade, with a \$50 fee and \$10 publishing cost, f_{cp} is \$1,350, above a \$1,000 collateral cap.

It is about 2.4 times f^*, and grows with window and volatility (Figure 3). Since it must be locked as collateral, the design is feasible only where this curve lies below the collateral line. Checking every window, f_{cp} fits the 1% limit up to 59 blocks (11.8 minutes) at the baseline volatility, so the 20-minute baseline is infeasible, while a 1 bp tolerance would allow 23.4 minutes (Table 17).

6.4 Q4. Contesting a disputed index

On chains where recent blocks can be replaced, one account slot (index) can end up with two valid commitments, the owner's and an attacker's, and a rule must decide which wins. By Proposition 6, a bond b deters if and only if $b \geq b_{\min} = \max\{0, ((1-a)V_a - \kappa)/a\}$, and works for sure if the owner can afford it and strictly prefers to answer, $K \geq b_{\min}$ and $V_o + K > c_r$. A race is an all-pay contest [52] and deters only if $\kappa \geq (1-a)V_a + a(V_a - V_o)^+$. In mechanism terms the bond is a payment the designer can scale, limited only by the owner's capital and its willingness to answer (automatic here, as $c_r \ll V_o$), while the race has nothing to tune. Across the owner's availability a and the value ratio V_a/V_o (Table 15), the race deters in only 2 of 12 cells, while the bond deters in all 11 cells where the owner can afford it, and both patterns hold across all 12 tested attack costs and capitals.

7 Discussion

The guarantees are easy to read as one stronger claim, so Table 4 states each precisely.

Property	What it means here
Deterrence	Weak: a speculator gains at most zero and, if indifferent, is assumed to stay out
Individual rationality	Honest users still take part, but without a fallback may fall slightly short of $v - \phi$
Credible fallback	Publishing is strictly better for a producer paid only fees and forfeits
Collusion-proofness	Joint gain over the best equilibrium is at most ε, a money amount (0.1 bp is \$1 on \$100,000), and $c(1-q^{k-1})$ more against waiting
Feasible window	Only for the chosen parameters, from 61.4 minutes at 2% daily volatility to 4.0 at 8%
Producers	One producer for the whole window; with many producers the coalition risk is overstated

Table 4. What each guarantee does and does not promise.

The forfeit from 14 calm days of real 20-minute ETH price changes, recentred to mean zero, is 0.169% with a 95% bootstrap interval of 0.136%–0.214%, against 0.212% for the lattice at the same volatility. This checks only the end-of-window distribution that Proposition 2 uses, not the block-by-block dynamics, and says nothing about stressed markets. All answers rest on a stylized model with fee-only producers, independent censoring, a fair lattice, risk neutrality and binding side contracts, though by Jensen's inequality a speculator deterred under risk neutrality stays deterred under risk aversion.

8 Conclusion and Future Work

Delaying authorization hands the user a free option. A forfeit alone must be 0.39% to 0.74% of a 20-minute trade (Q1). With a fallback it must not stay entirely with the user and producer, and sending it all outside, say by burning, is cheapest (Q2). At 20 minutes a burned 1.35% keeps collusion within tolerance, and the required forfeit fits 1% collateral only up to 11.8 minutes (Q3). An affordable, answered bond deters disputes that a race cannot (Q4). Future work should add order profits [25], known block schedules, an equilibrium-robust collusion guarantee and a screening menu for private types.

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Individual Contributions

This is a solo project. Reman Krishnan is the sole author and is responsible for the problem formulation, the model and its analysis, the propositions and proofs, the solver and its tests, the calibration and experiments, and this report.

Declaration of AI Use

Three AI models, Claude Opus 5.5, Sonnet 5.5 and ChatGPT 5.6 Luna, were used during this project. They helped organise and manage the GitHub repository, including its folder structure, documentation and reproduction scripts. They also helped me understand the game theory and mechanism design concepts that I needed to implement, and they assisted with the volatility calculations used to calibrate the model. When programming errors came up, I used them to find and fix the causes. Before submission, they carried out a thorough preliminary check of whether the implementation and the report met the requirements of the course project. They also helped me gather and verify relevant literature.

The models suggested corrections to the scope of a few claims, wording for the abstract and other sections, and helped in compiling the step by step layout of the proofs in the appendix. I checked every suggestion against the model, the code and the cited sources before including it, and I reran the test suite and the full pipeline after each change.

I chose the research questions, the model and the overall direction of the project, and I made the final decision on everything included in this report. I take full responsibility for its content, including any errors that remain.

A Proofs

This appendix proves the seven propositions one step at a time. Each step states one fact and the reason it holds. Throughout, $g_t = N(S_t/S_0 - 1)$ is the value at block t of executing the trade at the committed price, $\mathcal{F}_t$ is everything observed up to block t (prices and which attempts were included), and censoring is independent of prices.

Lemma A.1 (the lattice price is a martingale). $\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = S_t$, and therefore $\mathbb{E}[g_{t+1} \mid \mathcal{F}_t] = g_t$ and $\mathbb{E}[g_t] = g_0 = 0$ for every t.

Proof.

Step 1. From S_t the price moves to uS_t with probability p and to dS_t with probability $1-p$, independently of everything observed so far, so $\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = S_t\big(pu + (1-p)d\big)$.

Step 2. With $p = (1-d)/(u-d)$, $pu + (1-p)d = d + p(u-d) = d + (1-d) = 1$.

Step 3. Hence $\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = S_t$, and since g_t is an affine function of S_t with fixed coefficients, $\mathbb{E}[g_{t+1} \mid \mathcal{F}_t] = g_t$.

Step 4. By the tower property, $\mathbb{E}[g_t] = \mathbb{E}[g_{t-1}] = \cdots = g_0 = N(S_0/S_0 - 1) = 0$. □

Proof of Proposition 1.

Step 1. The terminal payoffs (user, producer) are $(g_t - \phi,\ \phi)$ for an authorization at block t, $(g_k - \phi,\ \phi - c)$ for a publication, and $(-(1-\rho_U)f,\ \rho_P f)$ for a drop (Figure 2).

Step 2. Self routing is $\rho = (1, 0, 0)$, so the drop payoff is $(-(1-1)f,\ 0 \cdot f) = (0, 0)$.

Step 3. The other terminal payoffs do not contain f at all.

Step 4. So every terminal payoff, and hence every expected payoff, every equilibrium and every coalition value, is the same for all f. □

Proof of Proposition 2. There is no fallback, $k \geq 2$ and $q = 0$, so every attempt is included and the user simply chooses when, if ever, to authorize. Let $\psi(x) = \max(x - \phi, -f)$ and $W_t = \mathbb{E}[\psi(g_{k-1}) \mid \mathcal{F}_t]$ for $t \leq k-1$.

Step 1. At block $k-1$, the last block with an authorization move, the user compares authorizing, worth $g_{k-1} - \phi$, with letting the commitment drop, worth $-f$. Hence $V_{k-1} = \psi(g_{k-1}) = W_{k-1}$.

Step 2. At a block t with $1 \leq t \leq k-2$, the user compares authorizing now, worth $g_t - \phi$, with waiting, worth $\mathbb{E}[V_{t+1} \mid \mathcal{F}_t]$. So $V_t = \max\big(g_t - \phi,\ \mathbb{E}[V_{t+1} \mid \mathcal{F}_t]\big)$.

Step 3. Suppose, as the induction hypothesis, that $V_{t+1} = W_{t+1}$. By the tower property, $\mathbb{E}[W_{t+1} \mid \mathcal{F}_t] = W_t$, so Step 2 gives $V_t = \max(g_t - \phi,\ W_t)$.

Step 4. ψ is convex, being the maximum of two affine functions. By Jensen's inequality and Lemma A.1,

$$W_t = \mathbb{E}[\psi(g_{k-1}) \mid \mathcal{F}_t] \ \geq\ \psi\big(\mathbb{E}[g_{k-1} \mid \mathcal{F}_t]\big) = \psi(g_t).$$

Step 5. By the definition of ψ, $\psi(g_t) \geq g_t - \phi$. With Step 4, $W_t \geq g_t - \phi$, so Step 3 gives $V_t = W_t$: waiting is always worth at least authorizing now.

Step 6. Steps 1 to 5 prove $V_t = W_t$ for $t = k-1, k-2, \ldots, 1$ by backward induction. Block 1 exists because $k \geq 2$. The value of committing is therefore $V_0 = \mathbb{E}[V_1] = \mathbb{E}[W_1] = \mathbb{E}[\psi(g_{k-1})]$, the first form.

Step 7. For any x, $\max(x, -f) = -f + (x+f)^+$. Putting $x = g_{k-1} - \phi$ gives $\psi(g_{k-1}) = -f + \big(g_{k-1} - (\phi - f)\big)^+$, and taking expectations gives the second form.

Step 8. For any number a, $(g-a)^+ - (a-g)^+ = g - a$. Taking expectations with Lemma A.1,

$\mathbb{E}[(g_{k-1}-a)^+] - \mathbb{E}[(a-g_{k-1})^+] = -a$.

Step 9. With $a = \phi - f$, Step 8 gives $\mathbb{E}[(g_{k-1}-a)^+] = \mathbb{E}[(a-g_{k-1})^+] - \phi + f$. Substituting into the second form, $V_0 = \mathbb{E}[(a-g_{k-1})^+] - \phi$, the third form.

Step 10. By the second form, $V_0 \le 0$ if and only if $\mathbb{E}[(g_{k-1} - (\phi - f))^+] \le f$, the deterrence condition.

Step 11. The lattice has finitely many nodes, so V_0 is a finite weighted sum of the functions $f \mapsto \psi(g)$, each continuous and non-increasing in f. Hence V_0 is continuous and non-increasing in f.

Step 12. Let $m = \min g_{k-1}$, which is finite. If $f \ge \phi - m$, then $g_{k-1} - \phi \ge m - \phi \ge -f$ at every node, so $\psi(g_{k-1}) = g_{k-1} - \phi$ and $V_0 = \mathbb{E}[g_{k-1}] - \phi = -\phi \le 0$.

Step 13. If $V_0(0) > 0$, the set $\{f \ge 0 : V_0(f) \le 0\}$ is non-empty by Step 12, closed by continuity, and an interval $[f^*, \infty)$ by monotonicity, so its least element f^* exists. □

Proof of Proposition 3. There is a fallback, the signature is handed over and there is no contestation.

Step 1. If the commitment reaches block k unauthorized, the producer gets $\phi - c$ by publishing and $\rho_P f$ by withholding. Neither number depends on the price or the history.

Step 2. So publishing is its unique best response at every such node if $\phi - c > \rho_P f$, withholding is if $\phi - c < \rho_P f$, and it is indifferent at equality.

Step 3. Suppose it publishes. Then the user's value on reaching block k unauthorized is $U_k = g_k - \phi$.

Step 4. Suppose that $U_{t+1} = g_{t+1} - \phi$ at block $t+1$, for some $1 \le t \le k-1$. Waiting at block t is worth $\mathbb{E}[U_{t+1} \mid \mathcal{F}_t] = g_t - \phi$ by Lemma A.1.

Step 5. Attempting at block t is worth $(1-q)(g_t - \phi) + q\,\mathbb{E}[U_{t+1} \mid \mathcal{F}_t] = (1-q)(g_t - \phi) + q(g_t - \phi) = g_t - \phi$.

Step 6. Both choices are worth $g_t - \phi$, so $U_t = g_t - \phi$, and by backward induction this holds at every block $t \ge 1$, for every $q \in [0,1]$.

Step 7. Hence $V_0 = \mathbb{E}[U_1] = \mathbb{E}[g_1] - \phi = -\phi$, and the forfeit is never charged, so this holds for every f.

Step 8. If instead the producer withholds, a commitment unauthorized at block k is dropped with user payoff $-(1-\rho_U)f$, exactly as without a fallback, so the two games coincide. □

Proof of Proposition 4. There is a fallback with the signature handed over, $k \ge 2$, $\sigma > 0$, $q < 1$, no contestation, $\rho_O = 0$ and $\phi - c > \rho_P f$.

Part 1: every equilibrium gives the pair at most 0.

Step 1. By Step 2 of the proof of Proposition 3, the strict inequality makes publishing the producer's unique best response at every unauthorized node of block k, in every equilibrium. So no commitment is ever dropped.

Step 2. Let τ be the block at which the trade executes, either the block of the first included authorization or k if the producer publishes. Whether $\tau \le t$ depends only on $\mathcal{F}_t$, so τ is a stopping time, and $\tau \le k$.

Step 3. The fee moves money inside the pair, so the joint payoff is $(g_t - \phi) + \phi = g_t$ for an authorization at block t and $(g_k - \phi) + (\phi - c) = g_k - c$ for a publication.

Step 4. Hence the expected joint payoff is $\mathbb{E}[g_\tau] - c\,\Pr(\text{publish}) \le \mathbb{E}[g_\tau]$.

Step 5. (g_t) is a martingale by Lemma A.1, since censoring is independent of prices, and τ is bounded by k, so the optional stopping theorem gives $\mathbb{E}[g_\tau] = g_0 = 0$.

Step 6. Every equilibrium therefore gives the pair an expected joint payoff of at most 0.

Part 2: a coalition gets $(1-q)\mathbb{E}[g_1^+] > 0$.

Step 7. With enforceable side payments, the pair can follow any joint strategy. Take this one: the user attempts at block 1 if and only if $g_1 > 0$ and never attempts otherwise, and the producer withholds at every node of block k.

Step 8. If $g_1 > 0$ and the attempt is included, which happens with probability $1-q$, the joint payoff is g_1.

Step 9. Otherwise the commitment is dropped, and the joint drop payoff is $-(1-\rho_U)f + \rho_P f = -(1-\rho_U-\rho_P)f = -\rho_O f = 0$.

Step 10. Since censoring is independent of the price, the coalition's expected joint payoff is $(1-q)\mathbb{E}[g_1 \mathbf{1}\{g_1 > 0\}] = (1-q)\mathbb{E}[g_1^+]$.

Step 11. This is positive. Block 1 is an authorization block because $k \geq 2$; $\sigma > 0$ gives $u > 1 > d$ and so $0 < p < 1$; the up-move gives $g_1 = N(u-1) > 0$ with probability p; and $1-q > 0$.

Step 12. So the coalition value exceeds every equilibrium joint payoff by at least $(1-q)\mathbb{E}[g_1^+] > 0$. With quasi-linear utilities this surplus can be split so that the user and the producer both gain strictly, so the design is not collusion-proof. □

The rotating-producer result of Section 4 is stated here.

Proposition 5 (Rotation disarms extortion)**.** *Let $k \geq 2$, $q = 0$, $c > 0$, $\rho_P > 0$, let there be a fallback, and let the user be honest with value $v > \phi$. Let each block be produced by a strategic entity with probability $\alpha \in (0,1]$, independently, and otherwise by an honest producer. If $\alpha^{k-1}\rho_P f < \phi$ the user loses nothing. If $\alpha^{k-1}\rho_P f > \phi$ the user's expected payoff is strictly below $v - \phi$. The extortion threshold is therefore $\phi/(\rho_P \alpha^{k-1})$, which is $\alpha^{-(k-1)}$ times the single-producer threshold ϕ/ρ_P. At equality the entity is indifferent at block 1, and the outcome depends on how this tie is broken.*

Proof of Proposition 5. Call the strategic entity E. Each block's producer is E with probability α, independently of everything else, and an honest producer otherwise, who includes every attempt and publishes at block k. Let W_t be E's expected payoff at the start of block t, before the producer of block t is drawn, while the user is still unauthorized.

Step 1. The honest user values execution at v whenever it happens, so attempting at every block weakly dominates waiting, and the user attempts at every block $1, \ldots, k-1$.

Step 2. At block k: with probability α, E produces and takes the better of $\phi - c$ (publish) and $\rho_P f$ (withhold); otherwise an honest producer publishes and E gets 0. So $W_k = \alpha \max(\phi - c,\ \rho_P f)$.

Step 3. At a block $t < k$: with probability $1-\alpha$ an honest producer includes the attempt and E gets 0. With probability α, E produces and gets ϕ by including or W_{t+1} by censoring. So $W_t = \alpha \max(\phi,\ W_{t+1})$, and E strictly prefers to censor at block t if and only if $W_{t+1} > \phi$.

Case 1: $\alpha^{k-1}\rho_P f > \phi$.

Step 4. Since $\alpha \leq 1$, $\rho_P f \geq \alpha^{k-1}\rho_P f > \phi > \phi - c$, so Step 2 gives $W_k = \alpha \rho_P f$.

Step 5. Suppose $W_{t+1} = \alpha^{k-t}\rho_P f$ for some $1 \leq t \leq k-1$. Since $k - t \leq k-1$ and $\alpha \leq 1$, $W_{t+1} \geq \alpha^{k-1}\rho_P f > \phi$, so Step 3 gives $W_t = \alpha W_{t+1} = \alpha^{k-t+1}\rho_P f$. By induction from Step 4, $W_{t+1} > \phi$ at every block $t = 1, \ldots, k-1$.

Step 6. Hence E censors at every block it produces and, by Step 4, withholds at block k.

Step 7. If E produces all k blocks, which has probability $\alpha^k > 0$, the commitment is dropped and the user gets $-(1-\rho_U)f \leq 0 < v - \phi$. Otherwise it gets $v - \phi$. Its expected payoff is therefore strictly below $v - \phi$.

Case 2: $\alpha^{k-1}\rho_P f < \phi$.

Step 8. Claim: $W_{t+1} \le \max(\alpha\phi,\ \alpha^{k-t}\rho_P f)$ for $t = k-1, \dots, 1$. For $t = k-1$ this is $W_k = \alpha \max(\phi - c, \rho_P f) \le \max(\alpha\phi, \alpha\rho_P f)$.

Step 9. If the claim holds at t, Step 3 gives $W_t \le \alpha \max\left(\phi,\ \alpha\phi,\ \alpha^{k-t}\rho_P f\right) = \max\left(\alpha\phi,\ \alpha^{k-t+1}\rho_P f\right)$, using $\alpha\phi \le \phi$. This is the claim at $t-1$.

Step 10. If $\alpha < 1$: at $t = 1$ the claim gives $W_2 \le \max(\alpha\phi,\ \alpha^{k-1}\rho_P f) < \phi$, because $\alpha\phi < \phi$ and by the case assumption. So E strictly prefers to include at block 1, an honest producer includes too, and the first attempt is included whoever produces block 1.

Step 11. If $\alpha = 1$: $W_k = \max(\phi - c,\ \rho_P f) < \phi$, because $c > 0$ and $\rho_P f = \alpha^{k-1}\rho_P f < \phi$. So E strictly prefers to include at block $k-1$, and the trade executes by block $k-1$ at the latest.

Step 12. In both sub-cases the commitment is never dropped. The honest user's value does not depend on when the trade executes, so it receives $v - \phi$ and loses nothing.

At $\alpha^{k-1}\rho_P f = \phi$ the entity is indifferent at block 1, and the outcome depends on how it breaks the tie. □

The bond-and-race result used in Q4 is stated here.

Proposition 6 (Bond and race)**.** *Deterrence means that attacking is not strictly profitable, and an attacker indifferent at a zero payoff is taken to abstain. (a) Let $a > 0$, $0 \le b \le K$ and $b_{\min} = \max\{0, ((1-a)V_a - \kappa)/a\}$. If a present owner answers, which it strictly prefers when $V_o + b > c_r$ and is indifferent about at equality, the attacker attacks if and only if $a(-\kappa - b) + (1-a)(V_a - \kappa) > 0$, so the bond deters if and only if $b \ge b_{\min}$. A deterring bond that the owner strictly prefers to answer therefore exists if and only if $K \ge b_{\min}$ and $V_o + K > c_r$; at $V_o + K = c_r$ the outcome depends on how the indifferent owner breaks its tie (the solver lets it answer). If $b > K$ the owner cannot answer. (b) Under a race, with the owner present the attacker's equilibrium payoff is $(V_a - V_o)^+$, so the race deters if and only if $\kappa \ge (1-a)V_a + a(V_a - V_o)^+$. The designer has no parameter that changes this threshold.*

Proof of Proposition 6, part (a): the bond.

Step 1. A present owner who answers keeps the index and wins the bond at cost c_r, a payoff of $b - c_r$, while not answering loses the index, a payoff of $-V_o$. So it strictly prefers to answer if and only if $b - c_r > -V_o$, that is, $V_o + b > c_r$, is indifferent at equality, and can answer only if it can post the bond, $b \le K$.

Step 2. If the owner is present and answers, which happens with probability a, the attacker pays its cost and loses the bond, $-\kappa - b$. Otherwise it takes the index, $V_a - \kappa$.

Step 3. The attacker's expected payoff from attacking is $a(-\kappa - b) + (1-a)(V_a - \kappa) = (1-a)V_a - \kappa - ab$, against 0 for not attacking.

Step 4. An attacker indifferent at 0 abstains, so the bond deters if and only if $(1-a)V_a - \kappa - ab \le 0$, that is, $b \ge ((1-a)V_a - \kappa)/a$. With $b \ge 0$, this is $b \ge b_{\min}$.

Step 5. A deterring bond that the owner strictly prefers to answer is a b with $b_{\min} \le b \le K$ and $V_o + b > c_r$, a guarantee that holds however an indifferent owner would break its tie. Since $V_o + b$ increases in b, such a b exists if and only if $b = K$ is one, that is, if and only if $K \ge b_{\min}$ and $V_o + K > c_r$.

Step 6. If $b > K$, the owner cannot post the bond, so it cannot answer. □

Part (b): the race.

Step 7. If the owner is absent, which happens with probability $1 - a$, the attacker takes the index unopposed, worth V_a before its cost.

Step 8. If the owner is present, the two pay for priority and the earlier inclusion wins, whatever the payments, which is a complete-information all-pay contest with values V_a and V_o.

Step 9. By Baye, Kovenock and de Vries [52], used here without a new proof, the player with the larger value earns the difference of the values in equilibrium and the other earns 0, so the attacker earns $(V_a - V_o)^+$.

Step 10. The attacker's expected payoff from attacking is $(1-a)V_a + a(V_a - V_o)^+ - \kappa$, so the race deters if and only if $\kappa \geq (1-a)V_a + a(V_a - V_o)^+$.

Step 11. This threshold depends only on a, V_a, V_o and κ, none of which the designer sets. □

The routing result used in Q3 is stated and proved here.

Proposition 7 (Burning is the cheapest routing)**.** *Fix k, $q \in [0,1]$, $0 < c < \phi$ and the price lattice, let the producer receive the signature and be allowed to publish it, and consider forfeit–routing pairs (f, ρ) that are strictly credible, $\rho_P f < \phi - c$. Then: (a) the coalition gain Δ depends on (f, ρ) only through $b = \rho_O f$, and is continuous and non-increasing in b; (b) the smallest burned forfeit F with $\Delta \leq \varepsilon$ exists, and a pair (f, ρ) meets (CP_ε) if and only if $\rho_O f \geq F$. Hence, if $F = 0$, a zero forfeit meets (CP_ε) under every routing. If $F > 0$, no routing with $\rho_O = 0$ meets it with a finite forfeit, and a routing with $\rho_O > 0$ meets it with the forfeit F/ρ_O, provided this forfeit is strictly credible, $\rho_P F/\rho_O < \phi - c$. The forfeit F/ρ_O is smallest at $\rho_O = 1$, when all of it leaves the pair, by burning or by payment to any passive outside recipient. For pool(n) with $n > 1$ it is $Fn/(n-1)$, strictly credible if and only if $n > 1 + F/(\phi - c)$, and pool(1) is producer routing, with $\rho_O = 0$; (c) under burn, any $f \geq c - \min g_k$ gives $\Delta = 0$.*

Proof.

Step 1. By strict credibility the producer publishes at every unauthorized node of block k in every SPE (Proposition 3), so nothing is dropped and, as in Steps 2 to 5 of the proof of Proposition 4, an equilibrium's joint payoff is $\mathbb{E}[g_\tau] - c\Pr(\text{publish}) = -c\Pr(\text{publish})$.

Step 2. Publication happens only if every attempt the user makes is censored. Attempting at every block makes this probability q^{k-1}, the smallest possible, and is an SPE because the user is indifferent among authorization times. So the best equilibrium joint payoff is $-c\,q^{k-1}$, which does not depend on f or ρ.

Step 3. For the coalition, the joint terminal payoffs are g_t for an authorization, $g_k - c$ for a publication, and $-(1-\rho_U)f + \rho_P f = -\rho_O f = -b$ for a drop. So J^*, the best joint payoff over all joint strategies, depends on (f, ρ) only through b.

Step 4. The lattice is finite, so there are finitely many pure joint strategies, and each gives an expected joint payoff that is affine and non-increasing in b. Their maximum $J^*(b)$ is therefore continuous and non-increasing, and so is $\Delta(b) = J^*(b) + c\,q^{k-1}$. This proves (a).

Step 5. By part (c), proved in Step 7 without using this step, $\Delta(b) = 0 \leq \varepsilon$ for $b \geq c - \min g_k$. With Step 4, the set $\{b \geq 0 : \Delta(b) \leq \varepsilon\}$ is non-empty, closed and an interval $[F, \infty)$, so its least element F exists. A burned forfeit has $b = f$, so F is the smallest burned forfeit with $\Delta \leq \varepsilon$, and a strictly credible pair (f, ρ) meets $(\mathrm{CP}_\varepsilon)$ if and only if $\rho_O f \geq F$.

Step 6. If $F = 0$, then $f = 0$ gives $\rho_O f = 0 \geq F$ under every routing. If $F > 0$ and $\rho_O = 0$, then $\rho_O f = 0 < F$ for every f. If $F > 0$ and $\rho_O > 0$, the condition reads $f \geq F/\rho_O$, so F/ρ_O is the smallest forfeit that works whenever it is itself strictly credible, and it is smallest at $\rho_O = 1$. Burning has $\rho_P = 0$ and is always strictly credible since $\phi > c$. Pool(n) with $n > 1$ has $\rho_P = 1/n$ and $\rho_O = 1 - 1/n > 0$, so its forfeit is $Fn/(n-1)$, and strict credibility $\rho_P f < \phi - c$ becomes $F/(n-1) < \phi - c$, that is, $n > 1 + F/(\phi - c)$. This proves (b).

Step 7. Under burn with $f \geq c - \min g_k$, at every node of block k we have $g_k - c \geq \min g_k - c \geq -f$,

so withholding is never better for the coalition than publishing. The coalition then faces the equilibrium problem of Steps 1 and 2, so $J^* = -c\,q^{k-1}$ and $\Delta = 0$. This proves (c). □

Solver recurrence. The lattice solver of Section 5 computes the user's value $V_t(j)$ at block t after j up-moves. At block k it is $g_k - \phi$ if the producer publishes and $-(1-\rho_U)f$ if the producer withholds or holds no signature, where the producer publishes when $\phi - c > \rho_P f$ (Proposition 3) and ties are handled as in Section 3. Writing $C_t(j) = pV_{t+1}(j+1) + (1-p)V_{t+1}(j)$ for the value of waiting, the value at an authorization block $1 \le t \le k-1$ is

$$V_t(j) = \max\big\{C_t(j),\ (1-q)\big(g_t(j) - \phi\big) + q\,C_t(j)\big\},$$

because an attempt succeeds with probability $1-q$ and otherwise the user simply continues. Finally $V_0 = pV_1(1) + (1-p)V_1(0)$. The coalition's value uses the same recurrence with the fee removed, $(1-q)g_t + qC_t$ for an attempt, and the terminal value $\max\{g_k - c,\ -\rho_O f\}$ when the producer holds Σ, since only the publishing cost and the share of the forfeit sent outside the pair leave the coalition.

B Design Choices at a Glance

Table 5 lists each modelling choice of the report, why it was made, why the obvious alternative was not taken, and what could be done instead or next. Section numbers point to where each choice is used.

C Supplementary Tables and Figures

This appendix collects material that supports the results in the main text and adds nothing to its claims. Money is in percent or basis points of the transaction value unless stated otherwise, and the window is $k = 100$ unless a table varies it.

Reading the tables and figures. Table 6 lists the checks of the solver. Figure 5 gives f^* in currency units, with $N = 100$ times the size. Table 7 gives the speculator's equilibrium value over the pooled period. Table 8 varies the fee, adds a fee paid at commitment, and compares the lattice with the closed form. Table 10 gives the extortion threshold under producer routing as a multiple of the single-producer threshold. In Table 11, each routing has 300 random designs, P1 is deterrence, P2 is strict credibility and P3 is exact collusion-proofness. Table 12 uses a fee of 5 bp and an honest surplus of 10 bp, and its forfeit grid is geometric, so the first loss appears at the first grid point above the threshold. Table 13 gives the designer's forfeit at $k = 41$ as a multiple of f^* under the 99% rule, the 95% rule and the exact minimiser. In Table 14, the last two columns give the speculator's value under a forfeit proportional to size calibrated at size one, and under a line fitted to the smallest deterring forfeits across the sizes. Table 15 varies the owner's availability a and the value ratio V_a/V_o. Each cell of Table 16 gives f^* without a fallback and then f_{cp} with a burned forfeit and a fallback, and a star marks f_{cp} above the 1% collateral limit. Table 17 varies the coalition-gain tolerance at the baseline volatility. Table 18 gives the best coalition gain over the equilibrium total in the fallback design. Table 19 uses a burned forfeit and no fallback. Table 20 prices an at-the-money American call over one day with 36 steps. It checks the solver against an independent textbook binomial loop, the method of Han et al., and does not use their per-pair volatilities. Tables 21 and 22 use the 14-day and two-month samples and report the Lo and MacKinlay z, which assumes constant volatility, with its heteroskedasticity-robust version.

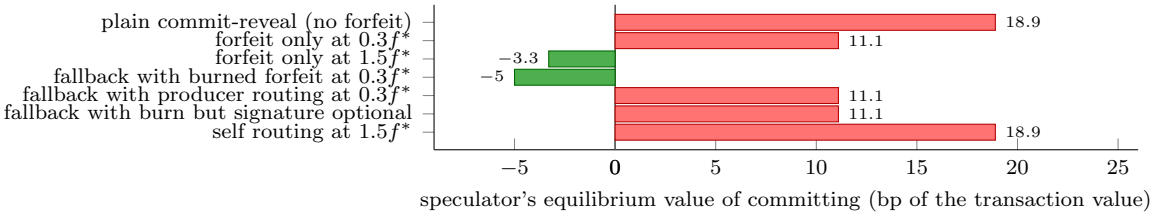


Figure 4. Speculator's value by design. Green bars are deterred.

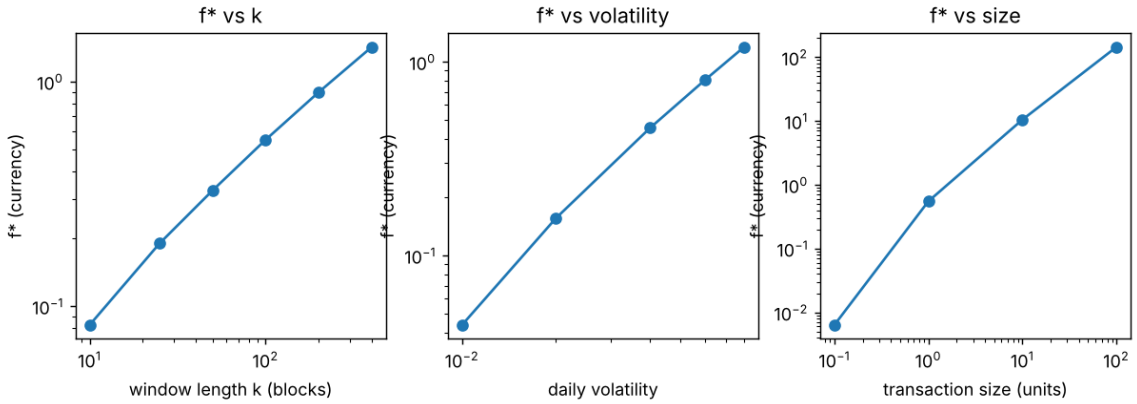


Figure 5. f^* in currency units against window, volatility and size.

Multiples of 100 and 300 in Table 21 are 20 minutes and one hour. In Figure 6, colour shows the smallest collateral fraction $\kappa_{\min}$, and "ok" marks the cells that are feasible at the assumed fraction.

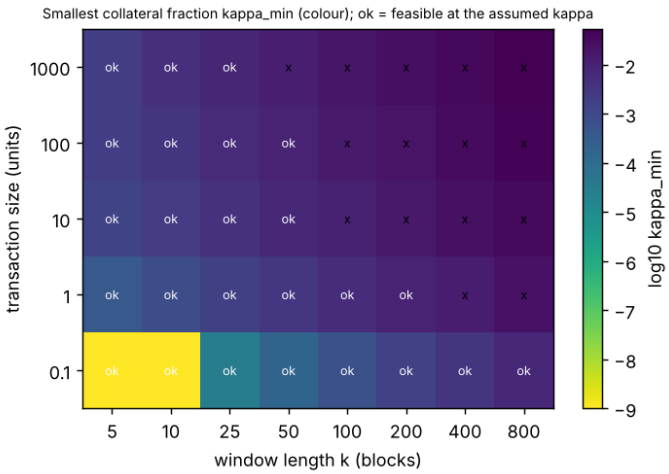


Figure 6. Smallest collateral fraction by window and size.

Choice (where)	Why	Why not the alternative	What else
Commitment with a producer fallback (Sec. 3)	Keeps the large post-quantum signature off the ledger unless the user stays silent	Always posting the signature removes the option but loses the saving; no fallback leaves the forfeit alone, which must be large (Q1)	Let any producer, or a watchtower, publish the signature
Binomial price lattice (Sec. 3)	A martingale, so the option has a clean value; exact backward induction; the same model Han et al. use	A continuous-time model has no exact finite solver; real paths are too short to calibrate a dynamic model	Fat tails and jumps, partly checked in Sec. 6
Risk neutrality and transferable money (Sec. 3)	Standard for payments; lets a coalition split any joint gain	Risk aversion only strengthens deterrence (Jensen's inequality, Sec. 8)	Risk-averse producers
Subgame perfection; deterrence in every pure equilibrium (Sec. 3)	Rules out threats a producer would not carry out (Appendix D)	Plain Nash equilibrium accepts such threats; one selected equilibrium could hide a profitable one	Mixed equilibria
Coalition gain against the best equilibrium (Sec. 3)	The most favourable honest benchmark; under a credible fallback it is the honest equilibrium	Against the waiting equilibrium every gain is larger by $c(1-q^{k-1})$ (Sec. 6)	An all-equilibria guarantee, or a reward for early authorization, each needing its own analysis
0.1 bp collusion tolerance (Q3)	A burned forfeit of $c - \min g_k$ suffices for exact collusion-proofness (Prop. 7) but is large	At the 20-minute baseline even 0.001 bp needs about 2%, above the 1% limit; short windows can do better	Tolerances from 0.001 to 1 bp (Table 17)
Routings self, burn, producer, pool(n) (Sec. 3)	Cover every place the money can go within the user–producer pair and outside it	Paying an outside party such as an index owner might let it provoke forfeits	Strategic recipients
Fee 5 bp, cost 1 bp, collateral 1%, window 20 minutes (Table 2)	Round illustrative values, since no deployed design exists to measure	Measured values do not yet exist	Sensitivity to fee, window and tolerance (Tables 8, 16, 17)
Daily ETH prices, 2017–2026 (Sec. 5)	Long, liquid history on a chain with 12-second blocks	Intraday samples are short and calm	Four assets, five periods, 12-second and one-minute checks
Independent rotating producers (Prop. 5)	Simplest model of a producer's share of blocks	Ethereum publishes its proposer schedule in advance, which this ignores	Known schedules and multi-block producers
Bond or race for disputed indices (Q4)	The two natural ways to settle a dispute without knowing who is honest	A fixed priority rule, such as the earlier commitment, may simply pick the attacker after a reorganization	Incomplete information about values
Private types with a 99% loss-reduction rule (Appendix D)	Shows that users with real need also exploit the delay	The exact minimiser is unstable because the loss is flat	A menu of (window, forfeit) pairs that screens types, as in mechanism design

Table 5. Modelling choices: why, why not and what else.

Check	Outcome
Hand-computed two-step game	$V_0 = 79/3$, and the solver gives 26.3333
Separate game-tree solver, history-dependent strategies	agrees under every uniform tie-breaking rule, all three models
Exhaustive enumeration of pure strategies	agrees on small games, including the bonded contest
Centipede game	the unique subgame-perfect one of 4 Nash equilibria
Textbook binomial call pricing (Han et al.), no fee or forfeit	matches to within 10^{-13}
All-pay contest	indifference conditions and simulation confirm it
Automated test suite, with deliberately injected bugs	all 196 tests pass, all 12 injected bugs caught

Table 6. Verification of the solver.

Design	f/f^*	speculator value (bp)	deterred?
plain commit-reveal (no forfeit)	0.0	18.9	No
forfeit only, small	0.3	11.1	No
forfeit only, large	1.5	−3.3	Yes
fallback + burn	0.3	−5.0	Yes
fallback + producer routing ($f >$ fee $- c$)	0.3	11.1	No
fallback + burn, σ optional	0.3	11.1	No
self routing	1.5	18.9	No

Table 7. Speculator's value by design (bp).

fee (bp)	fee at commitment (bp)	f^* (lattice)	f^* (closed form)
1	0	0.909%	0.917%
2.5	0	0.711%	0.716%
5	0	0.551%	0.554%
10	0	0.387%	0.388%
20	0	0.229%	0.229%
5	2.5	0.429%	
5	5	0.337%	
5	10	0.193%	

Table 8. Sensitivity of f^*/N to the fee.

Period	daily vol.	f^* (k=100)	plain commit-reveal, speculator value (bp)
Pooled 2017–2026	4.57%	0.56%	18.9
2018–2019	5.09%	0.65%	21.4
2020–2021	5.61%	0.74%	23.8
2022–2023	3.67%	0.41%	14.8
2024–2026	3.53%	0.39%	14.1

Table 9. Volatility and f^* by period.

stake share α	$k=4$	$k=8$	$k=12$	$k=20$
0.1	1,000	$>10^6$	$>10^6$	$>10^6$
0.25	64.0	16,384	$>10^6$	$>10^6$
0.5	8.0	128	2,048	524,288
0.75	2.4	7.5	23.7	237
0.9	1.4	2.1	3.2	7.4
1	1.0	1.0	1.0	1.0

Table 10. Extortion threshold under rotating producers.

routing	designs	P1 deters	P2 strict	P3 exact	all three
self	300	300	300	0	0
producer	300	241	199	6	0
mixed 50/50	300	238	211	9	0
pool(4)	300	244	215	15	0
burn	300	300	300	23	23

Table 11. Random search over designs.

routing	threshold fee/s_P	first loss on grid at $f=$	loss there
self	none	never	0
burn	none	never	0
pool(4)	0.200%	0.257%	0.357%
producer	0.050%	0.064%	0.164%

Table 12. Extortion by a single strategic producer.

q	99% rule/f^*	95% rule/f^*	exact minimiser/f^*	participation (99% rule)	loss at f=0 (bp)	loss at 99% rule (bp)
0.05	2.14	1.32	6.75	0.45	12.80	0.13
0.3	2.12	1.30	7.00	0.45	12.66	0.13
0.6	2.03	1.25	5.50	0.45	12.36	0.12

Table 13. Designer's forfeit with a private user type.

size	f^*/N, flat fee	f^*/N, fee $\propto$ size	speculator value, rule calibrated at size 1 (bp)	speculator value, line through the sizes (bp)
0.1	0.064%	0.551%	−31.1	−49.2
1	0.551%	0.551%	0.0	−4.9
10	1.048%	0.551%	3.8	−0.5
100	1.443%	0.551%	4.2	0.0

Table 14. Forfeits that grow with transaction size.

a	V_a/V_o	race deters?	attack cost to deter race	smallest bond	bond deters?
0.5	0.5	No	0.250	0.480	Yes
0.5	1	No	0.500	0.980	Yes
0.5	2	No	1.500	n/a	no (bond $> K$)
0.8	0.5	No	0.100	0.113	Yes
0.8	1	No	0.200	0.238	Yes
0.8	2	No	1.200	0.488	Yes
0.9	0.5	No	0.050	0.045	Yes
0.9	1	No	0.100	0.100	Yes
0.9	2	No	1.100	0.212	Yes
0.99	0.5	Yes	0.005	0.000	Yes
0.99	1	Yes (tie)	0.010	0.000	Yes
0.99	2	No	1.010	0.011	Yes

Table 15. Race against bond ($V_o = 1$, $\kappa = 0.01$, $c_r = 0.002$, $K = 1$).

k	daily vol. 2.00%	daily vol. 4.57%	daily vol. 8.00%
10	0.02 / 0.13	0.09 / 0.34	0.22 / 0.67
25	0.05 / 0.23	0.20 / 0.60	0.45 / 1.15*
50	0.09 / 0.35	0.33 / 0.91	0.74 / 1.69*
100	0.16 / 0.53	0.56 / 1.35*	1.20 / 2.50*
200	0.28 / 0.79	0.90 / 1.98*	1.88 / 3.66*

Table 16. f^* and f_{cp} by window and volatility.

tolerance (bp)	f_{cp} at k=100	largest k within 1%	minutes
1	0.91%	117	23.4
0.1	1.35%	59	11.8
0.01	1.69%	39	7.8
0.001	1.99%	29	5.8

Table 17. Sensitivity to the coalition-gain tolerance.

routing	producer publishes?	gain at $0.5f^*$ (bp)	gain at $1f^*$ (bp)	gain at $3f^*$ (bp)
burn	Yes	10.44	4.22	0.01
self	Yes	21.43	21.43	21.43
pool(4)	No	0.00	0.40	0.12
producer	No	1.66	8.51	21.22

Table 18. Coalition gain by routing (bp).

q	f^*/N	speculator value without a forfeit (bp)
0	0.551%	18.9
0.1	0.551%	18.9
0.3	0.549%	18.9
0.6	0.540%	18.7
0.9	0.475%	17.3

Table 19. f^* against the censoring probability.

daily vol.	Han method (%)	game solver (%)	|diff|
0.75%	0.2971	0.2971	8.1e-14
2.00%	0.7924	0.7924	1.1e-13
3.00%	1.1885	1.1885	9.0e-14
4.00%	1.5846	1.5846	8.3e-14
5.00%	1.9807	1.9807	2.8e-14
5.80%	2.2975	2.2975	1.2e-14

Table 20. Binomial premium against the game solver.

multiple m (12 s)	VR(m)	z	robust z	implied exponent
2	1.049	15.5	4.8	0.534
5	1.093	13.4	4.7	0.528
10	1.108	10.1	3.9	0.522
25	1.093	5.3	2.3	0.514
50	1.091	3.6	1.7	0.511
100	1.084	2.3	1.2	0.509
300	1.083	1.3	0.8	0.507

Table 21. Variance ratios of 12-second returns.

multiple m	VR(m)	z	robust z	implied exponent
2	1.006	1.8	0.3	0.504
5	1.001	0.1	0.0	0.500
10	0.978	-1.9	-0.4	0.495
30	1.000	0.0	0.0	0.500
60	1.005	0.2	0.1	0.501

Table 22. Variance ratios of one-minute returns.

asset	daily vol.	95% interval	f^* (k=100)
ETHUSDT	4.57%	4.23%–4.98%	0.56%
BTCUSDT	3.53%	3.25%–3.89%	0.39%
SOLUSDT	5.95%	5.38%–6.63%	0.80%
BNBUSDT	4.90%	4.28%–5.56%	0.61%

Table 23. Calibration and f^* by asset.

D Additional Results

Unknown user types. When a share π of users are speculators ($k = 41$, $f^* = 28.2$ bp), full deterrence is optimal in 36 of the 36 combinations of (q, π) tested, since an honest user is dropped with probability $q^{k-1} < 10^{-6}$ even at $q = 0.7$.

Private user types. Users may also differ in how much they genuinely need the trade, and this exposure may be private. Then moderately exposed users also exploit the delay, and the designer's forfeit, defined by the 99% loss-reduction rule below (not a 99% chance of deterrence), is 2.03 to 2.14 times f^* across the q tested, and 1.80 to 2.35 when the prior is varied. The prior and objective are illustrative, so this factor of about two is an illustration rather than a recommendation. Table 13 gives the values.

Nash equilibria and subgame perfection. The verification tools also show why the solution concept of Section 3 matters. With a burned forfeit of 50 in illustrative units and a fallback, only 4 of 25 pure Nash equilibria are subgame perfect, and the other 21 rest on a producer threat to withhold that it would never carry out.

Gaussian approximation. If g_{k-1} is roughly normal with standard deviation s, the put in Proposition 2 costs $a\Phi(a/s) + s\,n(a/s)$ with $a = \phi - f$, where Φ and n are the standard normal distribution function and density. This is a closed-form expression for the put, not for f^*; solving it equal to ϕ numerically gives an approximation to f^*, compared with the lattice in Table 8.

Fee paid at commitment. A fee paid at commitment lowers the speculator's value by a constant, and one of 5 bp lowers f^* from 0.551% to 0.34% (Table 8).

Forfeits that grow with size. Under the flat fee, a forfeit proportional to size and calibrated at size one leaves a speculator 4.2 bp at size 100. A forfeit fitted as a line across sizes over-penalizes the smallest trades 23 times (Table 14).

Loss function of the private-type model. A type commits if its best-response value V_{br} is at least zero, and V_{hon} is its value of authorizing at once. Its *option-exploitation benefit* is $X = V_{\text{br}} - V_{\text{hon}}$, the gain from timing its authorization instead of authorizing at once, and it bears a friction $F = (\theta - \phi) - V_{\text{hon}}$, while a deterred type loses its surplus $s(\theta) = \max(\theta - \phi, 0)$. The designer minimizes $L(f) = \mathbb{E}_\theta[\mathbf{1}_{\text{commit}}(\lambda X + \nu F) + \mathbf{1}_{\text{deterred}}\, s(\theta)]$ with weights λ and ν. X is a chosen proxy for harm to counterparties, not a measured loss. With $q = 0$ and $N = 1$, for example, $X = \mathbb{E}[g_\tau \mathbf{1}_{\text{exec}}] - (\theta - \phi + f)\Pr(\text{drop})$, where the first term is the counterparty's trading loss and the second is the user's own change in fees, forfeits and execution, so X equals neither the counterparty's loss nor a welfare loss in general. Because L is almost flat once exploitation vanishes, the designer's forfeit is defined as the smallest one that captures 99% of the reduction in L from a zero forfeit.

Prior of the private-type model. The prior puts weights 0.35, 0.20, 0.15, 0.12, 0.08, 0.06 and 0.04 on exposures of 0, 2, 5, 10, 20, 50 and 100 bp, and the loss weights are $\lambda = \nu = 1$.

Exact minimiser of the private-type loss. The exact minimiser rises from 1.0 to 12.2 times f^* as the number of steps in the window rises from 5 to 80 with the window's volatility held fixed. This is a joint sensitivity and not a pure refinement, because each extra step is also an extra authorization attempt with the same censoring probability q, so at $q = 0.6$ the chance that every attempt fails falls from $0.6^5 \approx 0.078$ to $0.6^{80} \approx 2 \times 10^{-18}$. A pure refinement would subdivide the price steps while keeping the authorization schedule, which is left for future work. The baseline at $k = 41$ is unaffected. The 99% rule is steadier. It is refined between coarse grid points, with the target taken from the coarse-grid minimum, and it gives 2.11 to 2.15 times f^* on the finest lattice. A forfeit grid with spacing $0.25f^*$ gives 2.25 and one with spacing $0.025f^*$ gives 2.13.

Intraday checks of the price model. The 12-second data are 14 days of one-second ETH closes, 9 October to 22 October 2026, resampled to 12-second blocks, which gives 100,700 overlapping 20-minute windows. This period is calm. The realized volatility per block is 0.0272% against 0.0538% implied by the full-history calibration, a daily equivalent of 2.3%, so the forfeits in the main text belong to a more volatile regime. The variance ratio is 1.05 at 24 seconds and between 1.08 and 1.11 at longer horizons (Table 21). A test that is robust to changing volatility finds it above one at the pointwise 5% level only up to about 5 minutes ($z = 4.8$ at 24 seconds and 2.3 at 5 minutes) and not at 20 minutes ($z = 1.2$), with no correction for the number of horizons. The point estimate is that square-root-of-time scaling understates the window's standard deviation by about 4%, which this sample cannot tell apart from zero. The forfeit under the empirical distribution is 0.169%, with a 95% interval of 0.136%–0.214% from 1,000 block-bootstrap resamples (seed 0, blocks twelve windows long). The one-minute data run from 2 August 2026 to 1 October 2026. There the interval 0.141%–0.178% excludes the lognormal value of 0.236% (ratio 0.67).

E Reproducing the Results

The repository https://github.com/reman-krishnan/cs6001-project contains the code, the price data and every result file. After installing the packages in `requirements.txt`, the single command `./run_all.sh` runs the tests, regenerates every result and rebuilds this report from the data in the repository (about 25 minutes). It uses only the frozen price data committed in `data/` and stops at the first failing step. The command `python scripts/compare_results.py` then checks every regenerated CSV file, staged or not, against the committed one, treating a changed NaN or a sign change of an infinity as a real difference. The pipeline was tested with Python 3.12 and TeX Live 2023, and `docs/REPRODUCING.md` lists the LaTeX packages needed. Setting `REFRESH_DATA=1` when running `scripts/run_assets.sh` downloads newer prices for the asset comparison and therefore changes those results. Results are written to `results/`, with the main calibration in `results/main/`. The individual steps are as follows.

- `python -m pytest` runs the automated tests.
- `python scripts/run_project.py` produces every table and figure for the main calibration.
- `python scripts/run_extras.py` produces the sensitivity and range analyses.
- `python scripts/run_course_core.py` produces the verification facts.
- `./scripts/run_periods.sh` and `./scripts/run_assets.sh` produce the period and asset comparisons.
- The one-minute checks and the 12-second checks come from the two commands below.

```
python scripts/run_empirical.py --data data/ETHUSDT_1m.csv
python scripts/run_empirical.py --data data/ETHUSDT_12s.csv \
  --window-minutes 20 --vr-multiples 2 5 10 25 50 100 300 \
  --n-boot 1000 --block-windows 12 --seed 0 --out results/intraday_12s
```

- `python scripts/make_report_assets.py` writes the tables and numbers used by this report.